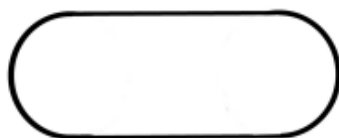
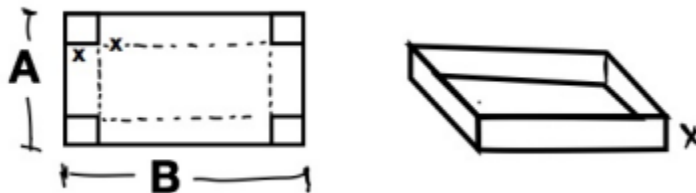


1. A farmer is going to use fence to make a pen shaped like cutting a circle in half and inserting a rectangle (like in the picture below).



If the pen is to have an area of 100π , how should he make the pen to minimize the amount of fence needed? Find all dimensions of the pen and sketch a picture of it.

2. For each $x > 0$, a triangle is formed with vertices $(0,0)$, $(x,0)$ and $(x, f(x))$ where $f(x) = \frac{10}{x^2 + 3x + 9}$ is the function given below. What is the value of x which results in the triangle of largest area? What is this largest area?
3. A rectangular box, without a top, is to be made by cutting squares out of corners of a sheet of cardboard and folding up the sides (like in the diagram below). Find the maximum volume of such a box from a sheet measuring 4 meters by 4 meters.



4. A rectangular box, without a top, is to be made by cutting squares out of corners of a sheet of cardboard and folding up the sides (like in the diagram below). Show that the maximum volume of such a box from a sheet measuring A meters by B meters occurs when the squares cut out have side lengths

$$x = \frac{(A+B) - \sqrt{(A+B)^2 - 3AB}}{6}$$



5. A cone-shaped drinking cup is made from a circular piece of paper of radius R by cutting out a sector and joining the edges of CA and CB . Find the angle θ that is needed in order to maximize the capacity of such a cup.

